VETERINARY MICROBIOLOGY · RAPID REVISION

Unit 3 — Microbial Biotechnology

All 10 syllabus topics on four working sheets. This unit is scored almost entirely on *process* — draw the steps in the right order, name the enzyme at each step, and state the one application. Every module here is built as a numbered sequence you can reproduce on the answer sheet.

PAPER	SYLLABUS TOPICS	REVISION TIME	COURSE
Paper I · Units 1, 2, 3 Theory 100 · Practical 60	10 of 10 covered	20 min full pass 8 min diagrams only	B.V.Sc & A.H. II Yr VCI MSVE · 3+2=5

Sheet	What it covers	Syllabus topics folded in
01	rDNA concept, the enzyme toolkit, vectors, the cloning cycle	t01–t02
02	Transformation, transfection, selection & screening; DNA libraries; probes	t03, t06, t07
03	PCR and its diagnostic variants; Southern, Northern & Western blotting	t04–t05
04	DNA sequencing, fingerprinting, bioinformatics, IPR & regulation	t08–t10

Core concepts & enzymes · Vectors & cloning · Amplification · Blotting & hybridisation · Sequencing & informatics · Exam trap & mnemonic

Recombinant DNA — Enzymes, Vectors & the Cloning Cycle

Gene cloning — the six steps, in order

MECHANISM

1. **1 · ISOLATE** Gene of interest — genomic DNA, cDNA or PCR
2. **2 · CUT** Same restriction enzyme on insert AND vector
3. **3 · LIGATE** T4 DNA ligase + ATP, 16 °C overnight → chimera
4. **4 · INTRODUCE** Transform competent host ($CaCl_2$ heat-shock / electro)
5. **5 · SELECT** Antibiotic marker + blue-white screening
6. **6 · EXPRESS** Induce, harvest & purify the recombinant protein

STICKY (COHESIVE) ENDS

5'-G AATTC-3'
3'-CTTAA G-5'

*Eco*RI · staggered cut · self-anneal easily → high ligation efficiency, directional cloning

BLUNT ENDS

5'-GTT AAC-3'
3'-CAA TTG-5'

*Hpa*I · straight cut · any blunt end joins any other → versatile but inefficient

The enzyme toolkit

TABLE

Enzyme	What it does & where it is used
Restriction endonuclease	Cuts dsDNA at a specific **palindromic recognition site** (4–8 bp). Bacterial defence against phage; the host protects itself by **methylation**. **Type II** is the one used in the lab — cuts within or close to the site, needs only Mg^{2+}
DNA ligase	Seals the nick — forms the **phosphodiester bond** between 3′-OH and 5′-phosphate. **T4 DNA ligase** works on sticky **and** blunt ends; *E. coli* ligase on sticky only
DNA polymerase I / Klenow	Fill-in of 5′ overhangs, labelling; Klenow lacks 5′→3′ exonuclease
Reverse transcriptase	RNA → cDNA. From AMV or M-MLV. The enzyme that makes cDNA libraries and RT-PCR possible
Alkaline phosphatase	Removes 5′-phosphate from the cut vector to **prevent self-ligation**
Terminal transferase	Adds homopolymer tails (poly-dA/dT) for homopolymer tailing
Nuclease S1	Degrades ssDNA — trims hairpins in cDNA synthesis
Taq polymerase	Thermostable, from *Thermus aquaticus*; **no 3′→5′ proofreading.** *Pfu* from *Pyrococcus furiosus* proofreads — use it for cloning
Lysozyme / proteinase K	Cell lysis and protein digestion during nucleic acid extraction

Naming rule ***Eco*RI** = **E** genus *Escherichia* · **co** species *coli* · **R** strain RY13 · **I** order of discovery. Genus letter capital and italic, species lower case italic, strain roman, number Roman numeral. ***Hin*dIII, *Bam*HI, *Pst*I, *Sma*I** follow the same rule.

Palindrome in DNA means the sequence reads the same 5′→3′ on **both** strands — GAATTC / GAATTC — not the same forwards and backwards on one strand. State it that way.

Vectors

TABLE

Vector	Insert capacity	Notes
Plasmid pBR322, pUC19	up to ~10 kb	Workhorse. pUC19 has a **polylinker/MCS inside *lacZ*** → blue-white screening
Bacteriophage λ	10–23 kb	High efficiency by *in vitro* packaging; **insertion** vs **replacement** vectors; used for genomic libraries
Cosmid	35–45 kb	Plasmid + λ ***cos*** **sites** — packaged like phage, replicates like a plasmid
Phagemid	~10 kb	Plasmid + f1 phage origin → produces ssDNA for sequencing/mutagenesis
BAC	100–300 kb	Based on the **F factor**; low copy, very stable — genome projects
YAC	200–2000 kb	Yeast **telomeres + centromere + ARS**; largest capacity, but chimerism is a problem
Expression vector	—	Adds a **strong inducible promoter** (*lac*, *tac*, T7), ribosome-binding site, terminator and often a **His-tag** for purification
Shuttle vector	—	Two origins → replicates in two hosts (*E. coli* + yeast/mammalian)
Ti plasmid	—	From *Agrobacterium tumefaciens* — the plant transformation vector
Viral vectors	—	Retro-, adeno-, adeno-associated, **poxvirus & herpesvirus** — the basis of **recombinant vector vaccines** in veterinary practice

Properties of an ideal vector **1** Origin of replication (autonomous) · **2** One or more **selectable markers** · **3** Unique restriction sites in a polylinker · **4** Small size and high copy number · **5** Easy to isolate · **6** Able to enter the host efficiently. Six points, six marks.

Scope of rDNA in veterinary science

RAPID-FIRE

Vaccines	**Subunit** (FMD VP1, *E. coli* K99 pilus), **vector** (poxvirus-vectored rabies for wildlife baits), **DNA vaccines, marker/DIVA vaccines** (gE-deleted IBR)
Diagnostics	PCR, real-time PCR, probes, recombinant-antigen ELISA, LAMP, biosensors

Restriction map arithmetic

MECHANISM

- **n cut sites in a linear** DNA → **n + 1** fragments.
- **n cut sites in a circular** DNA (plasmid) → **n** fragments. A single cut linearises the plasmid — one band.
- Fragment sizes must **add up to the total** — always check this in a problem.
- A **double digest** compared with the two single digests locates the sites relative to each other.
- **Partial digestion** gives extra, larger bands — used deliberately to build genomic libraries with overlapping fragments.

Sheet 01 · 60-second self-test

RECALL

Prompt	Answer
Molecular scissors	Restriction endonuclease
Molecular glue	DNA ligase
Bond made by ligase	Phosphodiester
Ligase that joins blunt ends	T4 DNA ligase
Enzyme making cDNA	Reverse transcriptase

Therapeutics	Recombinant bovine somatotropin, interferons, erythropoietin, insulin, monoclonal antibodies
Feed & industry	Recombinant phytase, cellulase, probiotics, single-cell protein, amino acids
Animal improvement	Marker-assisted selection, transgenic animals, gene knockouts, **CRISPR-Cas9** editing
Epidemiology	Molecular typing, outbreak tracing, phylogeny of field vs vaccine strains
Environment	Bioremediation, biofertilisers, biogas

Milestones: Berg 1972 first recombinant DNA · Cohen & Boyer 1973 first cloning in *E. coli* · Asilomar Conference 1975 · **Humulin 1982**, the first licensed rDNA product · Mullis 1983 PCR · Dolly 1996 · Human Genome Project 2003.

- Recognition site of **n** bp is expected once every **4^n** bp — a 4-cutter cuts every ~256 bp, a 6-cutter every ~4096 bp. That is why 6-cutters are used for cloning and 4-cutters for fingerprinting.
- **Isoschizomers** = different enzymes, same recognition site. **Neoschizomers** = same site, different cut position.

How to answer a gel/map question

1. Total the fragments of each single digest — they must agree.
2. Count sites: fragments − 1 for linear, = fragments for circular.
3. Place the enzyme-A sites on a line, then fit enzyme-B by seeing which A-fragment each double-digest band came from.
4. Label distances in kb and mark the scale.

Prompt	Answer
Prevents vector self-ligation	Alkaline phosphatase
Source of Taq polymerase	*Thermus aquaticus*
Proofreading polymerase	*Pfu*
Vector of largest capacity	YAC
Plasmid + *cos* site	Cosmid
Plant transformation vector	Ti plasmid of *A. tumefaciens*
First licensed rDNA product	Human insulin (Humulin, 1982)

02 Getting DNA In · Selecting It · Storing It in Libraries

t03, t06–t07

Introducing DNA into a host

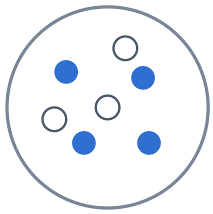
TABLE

Method	Host	How it works
Transformation	Bacteria, yeast	Uptake of **naked plasmid DNA** by a **competent** cell. Chemical: **ice-cold $CaCl_2$ then 42 °C heat shock for 90 s** — Ca^{2+} neutralises the phosphate charge and the shock opens transient pores
Electroporation	Bacteria & animal cells	High-voltage pulse creates reversible membrane pores; highest efficiency, no chemicals
Transfection	**Eukaryotic cells**	The same idea in animal cells, using **calcium phosphate co-precipitation, DEAE-dextran, liposomes (lipofection) or PEG. Stable** transfection = DNA integrates and persists; **transient** = expressed for a few days then lost
Transduction	Bacteria	Delivered by a bacteriophage — used for λ and cosmid libraries after *in vitro* packaging
Conjugation	Bacteria	Cell-to-cell via sex pilus; used for mobilisable vectors
Microinjection	Ova, embryos	Direct injection into the male pronucleus — the classic route for **transgenic animals**
Biolistics (gene gun)	Plants, skin	DNA-coated gold/tungsten microparticles fired into tissue; used for **DNA vaccination**
Viral vectors	Animal cells	Retro-, adeno-, AAV, pox — high efficiency, the basis of gene therapy and vectored vaccines

Say the difference cleanly: transformation is uptake of naked DNA by a bacterium; **transfection** is the same process in a eukaryotic cell; **transduction** is phage-mediated transfer. Examiners give a full mark for that single distinction.

Selection & screening of recombinants

MECHANISM



X-gal + IPTG plate + ampicillin

BLUE colony
Intact *lacZ* → β-galactosidase cleaves X-gal → blue dye = vector re-ligated, NO insert

WHITE colony
Insert interrupts *lacZ* — insertional inactivation = RECOMBINANT — pick this

Ampicillin first selects any transformant; X-gal then screens f

- **Selectable marker** — antibiotic resistance (amp^R, tet^R, kan^R); only cells carrying the plasmid grow.
- **Insertional inactivation** — the insert destroys a second marker. In pBR322, cloning into the *Bam*HI site kills tet^R, so recombinants are **amp^R tet^S**, found by **replica plating**.
- **Blue-white screening** — as drawn; the modern, single-plate version.
- **Direct screening of the insert** — colony PCR, restriction digestion of the miniprep, colony hybridisation with a labelled probe, and finally **sequencing** for proof.
- **Screening the product** — immunological screening with antibody against the expressed protein; complementation of a mutant host.

Genomic vs cDNA library

TABLE

Feature	Genomic library	cDNA library
Starting material	Total genomic DNA	**mRNA** of a chosen tissue
First step	Partial digestion with a restriction enzyme (or shearing)	**Reverse transcription** using oligo-dT primer
Contents	**All** the DNA — coding, non-coding, introns, promoters, regulatory regions	Only the **exons of genes expressed in that tissue at that moment**
Introns	Present	**Absent**

Nucleic acid hybridisation & probes

MECHANISM

- **Principle:** denatured single strands re-anneal only with a **complementary** sequence. A labelled known sequence (the **probe**) therefore finds its target in a complex mixture. **Specificity is controlled by stringency** — raise the temperature or lower the salt to allow only perfect matches.
- T_m rises with GC content, length and salt, and falls with formamide and mismatches. Hybridise about **20–25 °C below T_m**.
- **Probe types:** genomic DNA, cDNA, oligonucleotide (20–40 mer, synthesised from known sequence), RNA riboprobe.
- **Labels: radioactive** — ^{32}P, ^{35}S, ^{3}H, high sensitivity, autoradiography, but short half-life and a safety hazard; **non-radioactive — digoxigenin, biotin-avidin, fluorescein, alkaline phosphatase or HRP**, safe, stable, now standard.
- **Labelling methods:** nick translation, random primer (hexamer) labelling, end-labelling with T4 polynucleotide kinase, PCR labelling.

Feature	Genomic library	cDNA library
Size	Large — must cover the whole genome	Much smaller
Vector	λ, cosmid, BAC, YAC	Plasmid, λgt11, expression vector
Expression in *E. coli*	Eukaryotic gene **not** expressed correctly (introns cannot be spliced)	**Expressed** — this is why cDNA is used to make recombinant proteins
Chief use	Genome mapping, gene structure, regulatory elements	Protein production, expression profiling, gene identification

Library completeness Number of clones needed **N = ln(1 – P) / ln(1 – f)**, where P is the desired probability and f the fraction of the genome per insert. The practical point to state: a genomic library must contain enough **overlapping** clones that every sequence is represented at least once — that is why partial digestion is used.

- **Formats: Southern/Northern blot** (membrane-bound target) · **dot/slot blot** (crude, rapid) · **colony & plaque hybridisation** (library screening) · ***in situ* hybridisation and FISH** (target left inside the tissue or the chromosome) · **microarray** (thousands of probes on a chip, target labelled).
- **Veterinary use:** detecting non-cultivable agents, confirming PCR products, typing *Salmonella* and *E. coli* virulence genes, FISH for bacteria in tissue sections, chromosome painting.

Nucleic acid extraction & quantitation

RAPID-FIRE

Steps	Lyse (SDS + proteinase K / lysozyme) → deproteinise → precipitate → wash → dissolve in TE
Phenol-chloroform	Protein goes to the interphase; DNA stays in the upper aqueous layer
Precipitation	Chilled **absolute ethanol (2 vol) or isopropanol (0.6 vol) with 0.3 M sodium acetate**; wash in 70% ethanol
Plasmid isolation	**Alkaline lysis** (SDS + NaOH) — chromosomal DNA denatures irreversibly and is removed, supercoiled plasmid renatures; also boiling and column methods
RNA work	**TRIzol / guanidinium thiocyanate**; DEPC-treated water; RNase-free everything — RNase is the enemy
Quantitation	A_{260} **of 1.0 = 50 µg/ml dsDNA, 40 µg/ml** ssRNA, 33 µg/ml oligo
Purity	**A_{260}/A_{280} ≈ 1.8 for pure DNA, ≈ 2.0 for RNA.** Low = protein or phenol; A_{260}/A_{230} < 2.0 = salt/carbohydrate carryover
Other methods	Fluorimetry (Qubit, PicoGreen), gel comparison against a mass ladder, spectrophotometer/NanoDrop

Sheet 02 · 60-second self-test

RECALL

Prompt	Answer
Chemical making *E. coli* competent	Ice-cold $CaCl_2$ + 42 °C heat shock
DNA transfer into a eukaryotic cell	Transfection
DNA delivery by gold particles	Biolistics / gene gun
Substrate in blue-white screening	X-gal (inducer IPTG)
Colour of a recombinant colony	White
Gene inactivated by the insert	*lacZ*
Library without introns	cDNA library
Primer used to make cDNA	Oligo-dT
Commonest non-radioactive label	Digoxigenin / biotin
A_{260}/A_{280} of pure DNA	1.8
A_{260} = 1.0 corresponds to	50 µg/ml dsDNA
Plasmid isolation principle	Alkaline lysis

03 PCR & Its Variants · Blotting Techniques

t04–t05

PCR — the cycle, the components, the reading

MECHANISM

1 · DENATURATION
94–95 °C · 30 s
Hydrogen bonds break;
dsDNA → two single strands

2 · ANNEALING
50–60 °C · 30 s
Primers bind. The step that decides specificity — set 5 °C below T_m

3 · EXTENSION
72 °C · 1 min per kb
Taq adds dNTPs 5'→3';
final extension 72 °C / 5–10 min

REACTION MIX — name all seven
Template DNA · forward & reverse primers (18–25 nt)
dNTPs · **Taq polymerase** · **$MgCl_2$ (cofactor)**
10× buffer (KCl, Tris pH 8.3) · nuclease-free water

repeat 25–40 cycles → 2^n copies (exponential)

READING THE GEL: run the product on **agarose + ethidium bromide** beside a **100 bp ladder**. A band of the expected size = positive.
Always run a **positive control** (known DNA), a **negative/no-template control** (detects contamination) and an **internal control** (detects inhibitors). Nucleic acid migrates **cathode → anode** because the phosphate ba

PCR variants — what each one adds

TABLE

Variant	What it adds and where it is used in veterinary diagnosis
RT-PCR	A reverse-transcription step first → detects **RNA**. Essential for **FMD, PPR, rabies, influenza, IBD, bluetongue** and all RNA viruses
Real-time / qPCR	Product measured **during** the run by fluorescence — **SYBR Green** (non-specific, needs a melt curve) or **TaqMan hydrolysis probe** (specific). Gives **quantitation via the Ct value**, no gel, closed tube → far less contamination. Ct is **inversely** proportional to starting template

The three blots

TABLE

	Southern	Northern	Western
Target	**DNA**	**RNA**	**Protein**
Named after	**Edwin Southern,** 1975	A pun on Southern	A pun on Southern
Separation	Agarose gel	Denaturing agarose (formaldehyde)	**SDS-PAGE**

Variant	What it adds and where it is used in veterinary diagnosis
Nested PCR	Two rounds with an inner primer pair → very high sensitivity, but high contamination risk
Multiplex PCR	Several primer pairs in one tube → detects several pathogens or several virulence genes at once
Hot-start PCR	Polymerase blocked until the first denaturation → removes primer-dimers and non-specific products
Touchdown PCR	Annealing temperature stepped down cycle by cycle → specificity first, yield later
AP-PCR / RAPD	Single short arbitrary primer at low stringency → strain typing without sequence knowledge
RFLP-PCR	Amplicon digested with a restriction enzyme → pattern distinguishes species or vaccine from field strain
Inverse PCR	Amplifies **outward** from a known sequence into unknown flanking DNA
Digital PCR	Partitioned reaction → absolute quantitation with no standard curve
LAMP	Not strictly PCR — isothermal at ~65 °C in a water bath, 4–6 primers, read by turbidity or colour. **The field-level test** where no thermal cycler exists

Limitations to state: detects nucleic acid, not viability — a dead organism still gives a positive; inhibitors in faeces, blood and milk cause false negatives; carry-over amplicon causes false positives (control with separate rooms, aerosol-barrier tips, and **UNG/dUTP**); needs sequence knowledge; costly for routine herd screening.

	Southern	Northern	Western
Pre-treatment	Restriction digest, then **alkali denaturation** in the gel	Denaturant keeps RNA unfolded	SDS gives a uniform negative charge → separation by **size only**
Membrane	Nitrocellulose / nylon	Nylon	Nitrocellulose / PVDF
Transfer	Capillary, vacuum or electroblotting	Capillary	**Electroblotting**
Detected with	Labelled **DNA probe**	Labelled **DNA/RNA probe**	**Antibody** + enzyme-labelled conjugate
Tells you	Is the gene present? How many copies? Is it rearranged?	Is the gene **expressed**, and how much?	Is the **protein** present, and what size?
Vet use	RFLP typing, transgene confirmation, IS*900* in Johne's	Viral mRNA, expression studies	**Confirmatory test for BSE/scrapie (PrP^{Sc})**, EIA, bluetongue serology, FIV/BIV confirmation

Order to write the steps **Extract → separate → denature → transfer → immobilise (bake or UV cross-link) → block → probe/antibody → wash → detect (autoradiography or chemiluminescence).** Nine words, and they fit all three blots. **South-West blot** = DNA-binding protein; **Eastern blot** = post-translational modification; **dot blot** = no separation step at all.

Electrophoresis — AGE and SDS-PAGE

RAPID-FIRE

Principle	Charged molecules migrate in an electric field; separation by **size, charge and conformation**. DNA/RNA are uniformly negative → move to the **anode**
Agarose %	0.7% for large fragments (5–20 kb) · **1–1.5% routine PCR** · 2–3% for small products. Higher % resolves smaller fragments
Buffer	TAE (better resolution, low buffering) or TBE (better buffering, long runs)
Loading dye	Glycerol/sucrose to sink the sample + bromophenol blue & xylene cyanol as tracking dyes
Stain	**Ethidium bromide** — intercalates, orange under UV 300 nm; a **mutagen**. Safer: SYBR Safe, GelRed
Migration rule	Distance moved is **inversely proportional to $\log_{10}$ of the fragment size**; supercoiled runs faster than linear, which runs faster than nicked circular
SDS-PAGE	SDS denatures and coats protein with negative charge; **2-mercaptoethanol** breaks disulphide bonds; **Laemmli discontinuous system** — stacking gel (pH 6.8) concentrates, resolving gel (pH 8.8) separates by **molecular weight only**; stained with **Coomassie brilliant blue** or silver
Vet application	Plasmid profiling, protein profiling for strain differentiation, checking recombinant protein expression, preparing antigen for Western blot

Sheet 03 · 60-second self-test

RECALL

Prompt	Answer
Inventor of PCR	Kary Mullis, 1983 (Nobel 1993)
Three steps of a PCR cycle	Denature 94 °C · anneal 50–60 °C · extend 72 °C
Step deciding specificity	Annealing
Essential divalent cofactor	Mg^{2+} ($MgCl_2$)
Copies after n cycles	2^n
PCR for RNA viruses	RT-PCR
Value quantified in real-time PCR	Ct (threshold cycle)
Isothermal field amplification	LAMP (~65 °C)
Blot for DNA / RNA / protein	Southern / Northern / Western
Only blot using an antibody	Western
Confirmatory test for BSE	Western blot for PrP^{Sc}
Agarose gel stain	Ethidium bromide
Protein stain after PAGE	Coomassie brilliant blue

04 Sequencing, Fingerprinting, Bioinformatics & Regulation

DNA sequencing

MECHANISM

- **Sanger dideoxy chain-termination method (1977)** — the one to describe. Template + primer + DNA polymerase + all four dNTPs + a small proportion of one **dideoxynucleotide (ddNTP)**. The ddNTP lacks the **3′-OH**, so once it is incorporated no further phosphodiester bond can form and the chain stops. The result is a nested set of fragments differing by one base, resolved on a denaturing polyacrylamide gel or by **capillary electrophoresis**; the sequence is read from the smallest fragment upward. Modern automated version uses **four different fluorescent dyes on the four ddNTPs in a single tube**, read as an electropherogram.
- **Maxam-Gilbert chemical cleavage method** — base-specific chemical modification and cleavage; historically important, now obsolete because it uses hazardous chemicals.
- **Next-generation sequencing (NGS)** — massively parallel, millions of reads at once. **Illumina** (sequencing by synthesis, reversible terminators, short accurate reads), **Ion Torrent** (detects H^+ release, semiconductor), **PacBio** and **Oxford Nanopore** (single molecule, very long reads, portable). Applications: whole-genome sequencing of pathogens, metagenomics of the rumen and gut microbiome, virus discovery, transcriptomics (RNA-seq), outbreak tracing at single-nucleotide resolution.
- **Sequencing workflow terms:** read, coverage/depth, contig, scaffold, assembly, annotation.

The one-line answer if asked "why does a ddNTP terminate the chain?" — because it has no 3′-hydroxyl group, so the next nucleotide's 5′-phosphate has nothing to bond to.

DNA fingerprinting & molecular typing

TABLE

Technique	Basis & use
DNA fingerprinting	Developed by **Alec Jeffreys, 1984**. Based on **VNTRs (minisatellites)** and **STRs (microsatellites)** — tandem repeats whose copy number varies between individuals. Every individual except identical twins gives a unique pattern; the pattern is **inherited from both parents**
RFLP	Restriction digestion + Southern blot; classical but laborious
RAPD	Random primers, no sequence knowledge needed; quick but poorly reproducible
AFLP	Restriction + selective PCR amplification; high discrimination
PFGE	**Pulsed-field gel electrophoresis** — alternating field separates very large fragments; long the **gold standard for bacterial outbreak typing**
MLST	Sequences of 6–7 housekeeping genes → an allelic profile/sequence type; portable and comparable between laboratories
MLVA · spoligotyping	VNTR-based typing; spoligotyping for *M. bovis* strain tracing
Ribotyping · 16S rRNA	Identification and phylogeny of bacteria
SNP genotyping	Marker-assisted selection, parentage verification, disease-resistance loci

Veterinary applications Parentage and pedigree verification, **breed and individual identification of livestock**, meat speciation and adulteration detection, wildlife forensics and poaching cases, tracing the source of a food-borne outbreak, distinguishing **vaccine strain from field strain**, and marker-assisted selection.

Bioinformatics — databases and tools

RAPID-FIRE

Definition	Application of computational tools to store, retrieve, organise and analyse biological data — chiefly sequences and structures
The three primary nucleotide databases	**GenBank** (NCBI, USA) · **EMBL-ENA** (Europe) · **DDBJ** (Japan). They exchange data daily through the **INSDC** collaboration, so an accession number is valid in all three
Protein databases	**UniProt** (Swiss-Prot curated + TrEMBL automatic), PIR, **PDB** for 3-D structures
Specialised	PubMed (literature), OMIM, KEGG (pathways), Pfam (families), Rfam, GISAID (influenza & viral sequences)
BLAST	Basic Local Alignment Search Tool — the single most used tool. **BLASTn** nucleotide, **BLASTp** protein, **BLASTx** translated query vs protein DB. Output: score, **E-value (lower = more significant)**, % identity, query coverage
Alignment	**ClustalW/Omega, MUSCLE, MAFFT** for multiple alignment; pairwise by Needleman-Wunsch (global) and Smith-Waterman (local)
Phylogeny	**MEGA**; neighbour-joining, maximum likelihood, maximum parsimony; **bootstrap value** shows branch reliability

IPR, ethics & regulation

RAPID-FIRE

IPR	Legal rights over creations of the mind — **patent, copyright, trademark, trade secret, geographical indication, plant breeders' rights**
Patent	Exclusive right for **20 years** from filing, in exchange for full public disclosure. Three tests: **novelty, inventive step (non-obviousness) and industrial applicability**
Not patentable in India	Section 3 of the Patents Act — a mere discovery of a living thing or a naturally occurring substance, plants and animals in whole or in part, and traditional knowledge. **Processes and genetically engineered micro-organisms are patentable**
Landmark case	*Diamond v. Chakrabarty*, 1980 — a genetically engineered oil-degrading *Pseudomonas* was held patentable in the USA, opening biotechnology patenting
Budapest Treaty	Deposit of a micro-organism in a recognised depository (in India, **MTCC Chandigarh**) satisfies the disclosure requirement
International	**TRIPS** under WTO · **WIPO** · **Convention on Biological Diversity** (sovereign rights, access & benefit sharing) · **Cartagena Protocol on Biosafety** (transboundary movement of LMOs)
Indian law	Patents Act 1970 (amended 2005) · **Biological Diversity Act 2002** and NBA · Protection of Plant Varieties & Farmers' Rights Act 2001
Indian regulatory tiers for GMOs	Rules 1989 under the Environment Protection Act 1986: **IBSC** (institute level) → **RCGM** (DBT) → **GEAC** (MoEF, apex approval) → SBCC and DLC for monitoring
Animal ethics	**CPCSEA** and the **IAEC** clear all animal experiments; the **3 Rs — Replacement, Reduction, Refinement**; Prevention of Cruelty to Animals Act 1960
Biosafety levels	**BSL-1** to **BSL-4**; FMD and rabies work needs BSL-3, and BSL-4 is required for the most dangerous agents. Biosafety cabinets

Primer design	Primer3, Primer-BLAST; check for self-complementarity, GC 40–60%, similar T_m
Gene banks / repositories	*ex situ* conservation of germplasm, semen, embryos and DNA — **NBAGR Karnal** for animal genetic resources in India; **NBPGR** for plants; cryobanks for indigenous breeds

Common exam question: "How would you identify an unknown bacterial isolate using bioinformatics?" — amplify and sequence the **16S rRNA gene** → clean and assemble the read → **BLASTn against GenBank** → accept the top hit only if identity is **>97–99%** with good query coverage → confirm by building a phylogenetic tree with reference sequences in MEGA with bootstrap support.

	Class I, II and III
Ethical concerns to list	Animal welfare in transgenesis, biosafety and escape of GMOs, biopiracy of indigenous germplasm, antibiotic-resistance marker genes, food safety and labelling of GM feed, dual use and bioterrorism, equitable access

Sheet 04 · 60-second self-test

RECALL

Chain-termination sequencing	Sanger, 1977 — ddNTP lacks 3′-OH	**Chemical sequencing method**	Maxam-Gilbert
Sequencing by H^+ detection	Ion Torrent	**Longest-read sequencing**	PacBio / Oxford Nanopore
DNA fingerprinting discovered by	Alec Jeffreys, 1984	**Basis of fingerprinting**	VNTR / STR tandem repeats
Separates very large DNA fragments	PFGE	**Typing by housekeeping-gene sequences**	MLST
Three primary sequence databases	GenBank · EMBL-ENA · DDBJ	**Most-used sequence search tool**	BLAST — judge by E-value
Software for phylogenetic trees	MEGA	**Identity cut-off for species by 16S**	>97%
Term of a patent	20 years from filing	**Apex GMO regulator in India**	GEAC
Clears animal experiments in India	CPCSEA / IAEC	**The three Rs**	Replacement, Reduction, Refinement

`10 / 10 syllabus topics` `Paper I · Unit 3` `Full pass ≈ 20 min` `Night-before: sheet 03 PCR diagram + blot table + all self-tests`